Slide 1

Hello and thank you for being here.

Today, I am presenting a Project developed to provide Low-cost water level monitoring for municipal water management

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As a member of the AU^2 team working in partnership with the Augusta Utilities Department we focus much of our effort on improving and updating methods for monitoring our resources.

Upon discovery of a need for updated rate models in the Augusta area

We began our journey into water level monitoring systems with some **research and evaluation** of commercial options with **specific interest in their limitations.**

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High upfront costs additional maintenance, Replacement parts, and service contracts increase expense and limit scalability

Data transmission is often Bluetooth requiring manned operation for data retrieval

Systems that rely on main power or frequent battery changes can be *unreliable* in remote or off-grid installations.

Commercial systems utilize proprietary software often this means there is limited customization with fixed reporting formats or update intervals that may not be suitable for Municipalities needs.

If a supplier discontinues a product line or cloud service, municipalities can be left without updates or spare parts.

Because of these limitations we started looking for a different way to do this…

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Next we looked for **research publications** on monitoring water levels and found an **IEEE publication from 2018.**

This paper described a **promising experiment** that **utilized a Simple PVC pipe and loadcell** to obtain high resolution depth measurements and remote data transmission.

Unfortunately experimentation was conducted exclusively in a climate controlled laboratory.

Did not account for environmental conditions that impact accuracy of measurements.

The equipment used was cost prohibitive, not portable or suitable for extended outdoor use due to high power requirements.

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In conclusion of our research we compiled a list of goals for our **improved water level monitoring system H2 Float**.

Refer directly to the image

To Achieve these goals we will use a submerged pipe attached at the bottom to a loadcell. It measures buoyant force

We propose a device capable of **real time remote water level monitoring** that produces **high resolution measurements** with **low power requirements**.

We want a **low cost of production** with ease of **adaptability to site specific requirements** (water level variation and mounting options) to **ensure increased fielding** is possible.

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The **design basis of this submerged pipe** relies on **Archimedes' Principle**,

which states that the buoyant force acting on a submerged object is equal to the weight of the displaced fluid.

In this system, the **pipe is submerged in water**, and as the water level increases or decreases,

the **submerged volume of the pipe and therefor the buoyant force will likewise increase and decrease**.

Since the depth is directly proportional to the buoyant force we measure using the loadcell

We can translate the force measurements directly into the desired depth measurements

How do we calibrate this?

With PVC having standard sizes we had to consider the impact of the radius for each standard size.

***Resolution** relies heavily on sensitivity to changes in depth. A **smaller radius** responds faster with lower resolution and increased flexibility under strain. A **larger radius** gives a greater change in force for a small change in depth improving sensitivity, resolution and rigidity however too large of a radius increases costs.*

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First we submerge the pipe to the desired Zeroed depth position

this is usually around the midpoint of the pipe.

Next we run a 2 point calibration program this **averages 100 measurements each** at **tare and with a 750g weight**.

These values are then used to perform a linear fit.

Which establishes a constant zeroed depth reference from which all depth measurements up and down are based.

The **code** can **easily be adapted** to provide **measurements in any system** the user wants.

A common choice is to show values in **feet above Sea level**.

You may ask how are we certain our calibration is accurate?

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We tested the accuracy of our calibration by taking **measurements at known depths** and **compared them to the measured values from the device**.

For Perfect Calibration we expect a straight line with a slope of 1

As you can see in this graph **our** straight line fit is **accurate to 0.13% systematic error** which demonstrates the high precision and reliable linear response of the device.

Error bars are smaller than the markers

After confirming the **device provides accurate measurements** we decided to **see how it works in the real world...**

Data for the error bars were obtained independently from depth testing measurements using 2 devices calibrated to zeroed positions taking measurements every 15 minutes for a period of more than a week. The fluctuation from zero position remained inside of +4mm and -1mm.

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Here you see our prototype. It is mounted to a dock in a pond in Columbia county.

Achievements of our initial deployment

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Speak about the picture

you this device is larger than our prototype with a self contained and the size is customizable and still compact

surrounded by a corrugated and perforated pipe

allowing water to move freely through

this provides rigidity and protection for the device inside......debris and rough water conditions

There is a lithium Polymer battery inside the components box

You see the solar cell keeps it charged

There is wireless communication and

Sd card backup

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You See

MKR1310 microcontroller with Integrated LoRa transmitter

long range radio transmission - operates in remote areas **No cellular Required** offer remote monitoring.

Transmits Data to gateway

LoRa device **outside range of the gateway** the data can be sent through other devices in a **daisy chain** manner until it reaches a device that connects to the gateway.

Solar Cell Integration runs device on solar, charges battery and automatically switched between the 2

Sd card Backup

Require manual firmware updates meaning it is best for the location of device to be easily accessed.

As you will see shortly the device runs for long periods of time without need for updates or intervention ...the point still remains.

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Here is a diagram of how this is put together

Prev slide there is a microcontroller powered by solar with battery back up

data gets sent to gateway on device remotely in real time

Sd Card Not shown

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Address this picture.

Because we have opted to use this version of the device in the Augusta area we have developed a Printed Circuit Board to **simplify production** and **reduce costs** while **improving reliability** and **enhancing durability**

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Dashboard for analysis

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Time and date stamp

Set currently to take measurements on a 15 minute interval.

This is easily adjusted to desired intervals of the user.

Depth measurements can be customized to the system of choice.

Battery voltages are always recorded

Indicator of the status of low power programming.

ensure the battery is not nearing the end of its life cycle

Provides early warning for when a battery change that may be needed

How are we using these?

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Here is a map of the locations where we have this sensor deployed currently.

As we continue this project we would like to deploy more devices along the canal and in the waterways that feed lake Olmstead.

We would also like to deploy devices in the savannah river above and below the canal headgates.

This will provide us with data needed to adjust current rate models for this part of the Augusta area.

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Picture Talk about versatility and Placement

Substantial debris build up

The device to the left of H2float is a ultra sonic sensor.

It produced nearly identical data until the debris built up along the water surface.

In this image the ultra Sonic sensor is reading debris build up

FERC- Federal Energy Regulatory Commission

If you notice this image there is a spillway

2 more up stream

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5 weeks

7/11 there was a period of time where all three spillways ahead of this device closed and did not reopen

7/15 The sudden drop in the water level this was in preparation for an expected influx of water

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When placing a device we carefully consider the known water level variations

Each

Accessing this device requires a boat launch so it’s important that this device maintains battery level after several days of low solar

device needs to have enough range

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Data 3 weeks

We did a firmware update and increased the **<u>SAMPLING RATE</u>**

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What would you expect to pay for a device like this

Increased Location monitoring